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16

Deterministic Lower Bounds

The aim of this chapter is to provide a general technique for computing lower bounds for decision problems over the reals. These bounds come from upper bounds on the number of connected components of real algebraic sets. Besides being interesting in their own right, results proved here are used in later chapters.

16.1 A Geometric Upper Bound

The main result of this section, Theorem 1, bounds the number of connected components of a real algebraic set by an expression that is exponential in the dimension of the ambient space and polynomial in the degrees of the defining polynomials. In order to prove this fact we begin with a very special case, namely, a smooth compact hypersurface. Many of the technical notions used in this section have been defined in Chapters 8, 9, and 10.

Proposition 1 *If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a polynomial of degree d , δ is a regular value of f , and the hypersurface $H_\delta = \{x \mid f(x) = \delta\}$ is compact, then H_δ has at most $(d(d-1)^{n-1})/2$ connected components.*

The proof relies on some results of differential topology. Before proving them, we give an outline of the proof for motivation.

Outline of the Proof of Proposition 1. Pick an orthogonal projection $L : \mathbb{R}^n \rightarrow V$, where V is a one-dimensional subspace of $\mathbb{R}^n$ so that $L|_{H_\delta}$ is a Morse function; that is, any critical point of $L|_{H_\delta}$ is nondegenerate.

If we change coordinates to make V the subspace $(0, 0, \dots, x_n)$, and L the last coordinate function, then (Lagrange multipliers say that) the equations for the critical points are

$$f - \delta = 0, \quad \frac{\partial f}{\partial x_1} = 0, \dots, \frac{\partial f}{\partial x_{n-1}} = 0.$$

Nondegeneracy asserts that the gradients of these n functions are independent. These polynomials have degree d for f and $d - 1$ for each partial derivative. Bézout's theorem says then that they have at most $d(d - 1)^{n-1}$ isolated roots even in $\mathbb{C}^n$. Since each connected component has a max and a min, the connected components are in number at most half the critical points.

To complete the argument along these lines requires several lemmas. First we include a proof that a projection L may be found. We begin by considering

$$g : H_\delta \rightarrow S^{n-1} \\ x \rightarrow \frac{\text{grad } f(x)}{\|\text{grad } f(x)\|}.$$

Since δ is a regular value of f , $\text{grad } f(x) \neq 0$ for any x in H_δ and g is well-defined.

Lemma 1 *The following assertions hold.*

- (i) *Almost all $y \in S^{n-1}$ are regular values of g .*
- (ii) *If y is a regular value of g , then the projection π_y on the one-dimensional subspace spanned by y is a Morse function when restricted to H_δ .*

Proof. Part (i) is a special case of a basic result of differential topology, *Sard's Theorem*, which asserts that the measure of the set of critical values of a smooth mapping between differentiable manifolds is zero. We may see it in the case at hand very simply. First, we can cover H_δ by a finite set of charts (see Definition 2 of Chapter 10), each one being an $(n - 1)$ -dimensional closed cube, and prove our statement for these cubes. Thus, in the following we assume H_δ is a closed cube.

If x is an element of the critical set of g , then the image of $Dg(x)$ is a proper subspace H of the tangent space $T_{g(x)}$ of S^{n-1} at $g(x)$. Consider the composition $\tau (= \tau_x)$,

$$H_\delta \xrightarrow{g} S^{n-1} \xrightarrow{\pi} H^\perp$$

of g with the projection π on the orthogonal complement $H^\perp$ of H in $T_{g(x)}$. Since π restricted to H is zero, so is $D\pi$ and one has that $D\tau(x) = (D\pi)(g(x)) \circ Dg(x)$ is zero as a linear map.

Let us consider $\varepsilon > 0$ and $y \in B_\varepsilon(x)$. By the mean value theorem one has that

$$d(\tau(x), \tau(y)) \leq \varepsilon \max_{\zeta \in B_\varepsilon(x)} \|D\tau(\zeta)\|.$$

Since the derivative of τ at x is 0 and H_δ is a closed cube, by the mean value theorem applied now to $D\tau$, we have that there is an $\varepsilon_0 > 0$ and a constant $c > 0$,

which do not depend on the critical point x , such that $\max_{\zeta \in B_\varepsilon(x)} \|D\tau(\zeta)\| \leq c\varepsilon$ for all $0 < \varepsilon < \varepsilon_0$. Therefore

$$d(\tau(x), \tau(y)) \leq c\varepsilon^2$$

and thus, $\tau(B_\varepsilon(x)) \subseteq B_{c\varepsilon^2}(g(x))$ where the last ball is in $H^\perp$. We deduce that $g(B_\varepsilon(x)) \subseteq \pi^{-1}(B_{c\varepsilon^2}(\tau(x)))$, that is, that the image by g of the ε ball around x is contained in the $c\varepsilon^2$ neighborhood of the sphere S tangent to $H^\perp$ through $g(x)$.

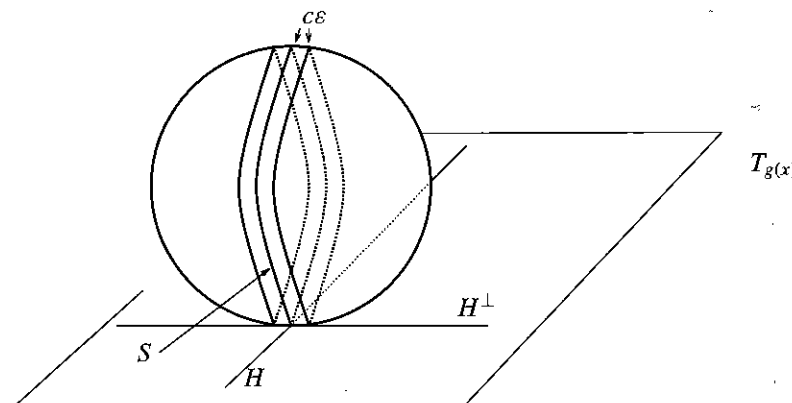


Figure A The strip around S and the image by g of $B_\varepsilon(x)$ contained on it.

By a similar argument, there are constants $c' > 0$ and $\varepsilon_1 > 0$ such that $g(B_\varepsilon(x))$ is included in $\pi_2^{-1}(B_{c'\varepsilon}(g(x)))$ for all $0 < \varepsilon < \varepsilon_1$, where $\pi_2 : S^{n-1} \rightarrow H$. Thus, for $0 < \varepsilon < \min\{\varepsilon_0, \varepsilon_1\}$ and any critical point x of g in H_δ ,

$$\text{Vol}(g(B_\varepsilon(x))) \leq (c\varepsilon^2)^{\dim H^\perp} \cdot (c'\varepsilon)^{\dim H} \\ \leq c_0 \varepsilon^n$$

for another constant c_0 , since $\dim H^\perp + \dim H = n - 1$ and $\dim H^\perp \geq 1$.

Let L be the length of a side of the cube H_δ and $k \in \mathbb{N}$. Then H_δ can be covered by k^{n-1} subcubes of side L/k . We see that the volume of the image of the critical points of g in H_δ is at most $(c_0 L^n n^{n/2})/k$ for every $k \in \mathbb{N}$ large enough and hence is zero.

For Part (ii), let $y \in S^{n-1}$ be a regular value of g . By rotation we may assume $y = (0, \dots, 0, 1)$. As previously computed, the critical points of the projection are then the zeros of $f - \delta = 0$, $\partial f / \partial x_1 = 0, \dots, \partial f / \partial x_{n-1} = 0$; that is, p is a critical point if and only if $\text{grad } f(p) / \|\text{grad } f(p)\| = y$. The tangent space of H_δ at p is given by $\sum (\partial f_i / \partial x_i) v_i = 0$, thus at a critical point by $v_n = 0$.

Now by the product rule for derivatives the derivative of $\text{grad } f / \|\text{grad } f\|$ at a critical point p applied to a tangent vector v is a multiple of $(D \text{grad } f)(p)(v)$ minus a multiple of y . Since y is a regular value this linear map is an isomorphism from the tangent space of H_δ at p to the tangent space of S^{n-1} at y . But the tangent space of S^{n-1} at y is orthogonal to y and therefore $\pi_{y^\perp}(D \text{grad } f)(p)$ is

an isomorphism on the tangent space to H_δ at p , where $\pi_{y^\perp}$ is the projection on the orthogonal complement of y . Once again using that the tangent space to H_δ at p is defined by $v_n = 0$ it follows that the first $n - 1$ columns of $(D \text{grad } f)(p)$ projected into the tangent space of H_δ at p are independent vectors. Since the tangent space to H_δ at p is defined by the gradient of $f - \delta$ and the first $n - 1$ columns of $(D \text{grad } f)(p)$ are the gradient vectors of $\partial f / \partial x_1, \dots, \partial f / \partial x_{n-1}$, the gradient vectors of $f - \delta, \partial f / \partial x_1, \partial f / \partial x_2, \dots, \partial f / \partial x_{n-1}$ are independent at a point p where $f - \delta, \partial f / \partial x_1, \partial f / \partial x_2, \dots, \partial f / \partial x_{n-1}$ are all zero. Thus the critical points of $\pi_y|_{H_\delta}$ are nondegenerate. $\square$

Lemma 2 Let $g_1, \dots, g_n : \mathbb{R}^n \rightarrow \mathbb{R}$ be polynomials with degree $g_i \leq d_i$ that have k common zeros. If at each of these zeros the gradients $\text{grad } g_i$, for $i = 1, \dots, n$, are independent, then $k \leq \prod_{i=1}^n d_i$.

Proof. Consider the polynomials g_i as polynomials of complex variables $g_i : \mathbb{C}^n \rightarrow \mathbb{C}$. Then the complex gradients are the same as the real gradients, hence independent. Thus, by the inverse function theorem, the roots are isolated in $\mathbb{C}^n$ and by Bézout's theorem there are at most $\prod_{i=1}^n d_i$ isolated roots. $\square$

Now we may finish the proof of Proposition 1.

Proof of Proposition 1. Apply Lemma 2 to $f - \delta, \partial f / \partial x_1, \dots, \partial f / \partial x_{n-1}$ (after rotating the regular value y in Lemma 1 to be equal to $(0, 0, \dots, 0, 1)$). $\square$

This finishes the proof of Proposition 1. We need Sard's Theorem in a different but also elementary setting shortly. The version we use, as well as its proof, is given in the next proposition.

Proposition 2 If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a polynomial, then f has only finitely many critical values.

Proof. The critical points are zeros of $\text{grad } f : \mathbb{R}^n \rightarrow \mathbb{R}^n$. We may complexify and consider $\text{grad } f : \mathbb{C}^n \rightarrow \mathbb{C}^n$. The zeros of the real gradient are zeros of the complex gradient. The zeros of the complex gradient are either all of $\mathbb{C}^n$, and f is constant, or there are finitely many connected components for the homogenized equations by Bézout's theorem and f is constant on each of them. Thus there are only finitely many critical values. $\square$

Note that in the proof of the preceding proposition we needed to complexify first. Bézout's Theorem is not true over $\mathbb{R}$ as can be seen in the following example.

Example 1 Let $f_1, \dots, f_n : \mathbb{R}^n \rightarrow \mathbb{R}$ be real polynomials with $\deg(f_i) = d_i$ such that the set of their common zeros has $\prod_{i=1}^n d_i$ isolated points. For instance, $f_i = \prod_{j=1}^{d_i} (x_i - a_{ij})$ with $a_{ij} \neq a_{ik}$ for $j \neq k$.

Then $f = f_1^2 + \dots + f_n^2 = 0$ has degree $d = \max_i(2d_i)$. Now consider the set of polynomial equations

$$\left\{ \begin{array}{l} f = 0 \\ x_{n+1} = 0 \\ \vdots \\ x_{n+1} = 0 \end{array} \right\} n \text{ times.}$$

The zero set of this system has $\prod_{i=1}^n d_i$ connected components and the product of the degrees of the f_i is d .

Now we turn to our main theorem dealing with the more general case of a not necessarily compact or smooth set.

Theorem 1 Let $V \subset \mathbb{R}^n$ be defined by $f_1 = 0, \dots, f_p = 0$, where f_i are polynomials of degree at most d . Then the number of connected components of V is at most $d(2d - 1)^{n-1}$.

Proof. If we denote by $B(r)$ the ball centered at the origin with radius r , it suffices to prove this estimate for $V \cap B(r)$ for a sequence of r s that go to infinity, since any connected component of V must ultimately intersect $B(r)$ for large r and contributes at least one connected component to the intersection. Consider the function

$$F_\varepsilon(x_1, \dots, x_n) = f_1^2(x_1, \dots, x_n) + \dots + f_p^2(x_1, \dots, x_n) + \varepsilon^2(x_1^2 + x_2^2 + \dots + x_n^2).$$

For every $\varepsilon > 0$ almost every $\delta > 0$ satisfies that δ^2 is a regular value of F_ε . Let W_δ be defined by $W_\delta = \{(x_1, \dots, x_n) \mid F_\varepsilon(x_1, \dots, x_n) \leq \delta^2\}$ and $H_\delta = \{(x_1, \dots, x_n) \mid F_\varepsilon(x_1, \dots, x_n) = \delta^2\}$.

Note that $W_\delta \subseteq B(\delta/\varepsilon)$. Moreover, since δ is a regular value of F_ε , by the implicit function theorem W_δ is the closure of its interior and $H_\delta = \partial W_\delta$. See Figure B.

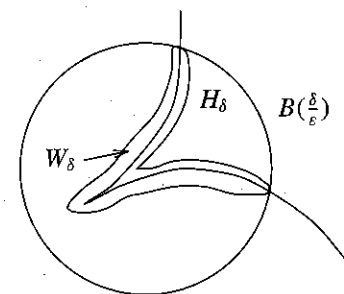


Figure B The sets W_δ and H_δ .

Lemma 3 If ε_n, δ_n converge monotonically to zero, δ_n/ε_n converges monotonically to r from above, and δ_n is a regular value of F_{ε_n} , then $W_{\delta_{n+1}} \subseteq W_{\delta_n}$ and $\cap_n W_{\delta_n} = V \cap B(r)$.

Proof. To see that $W_{\delta_{n+1}} \subseteq W_{\delta_n}$ note that

$$\frac{\sum f_i^2}{\delta_{n+1}^2} + \frac{\varepsilon_{n+1}^2}{\delta_{n+1}^2} \|x\|^2 \leq 1$$

implies

$$\frac{\sum f_i^2}{\delta_n^2} + \frac{\varepsilon_n^2}{\delta_n^2} \|x\|^2 \leq 1$$

since $\delta_{n+1} \leq \delta_n$ and $\varepsilon_n/\delta_n \leq \varepsilon_{n+1}/\delta_{n+1}$.

The other part of the statement follows from the facts: (i) $\sum f_i^2 \leq \delta_n^2$ for all $n \in \mathbb{N}$ implies that $f_i = 0$ for $i = 1, \dots, p$, and (ii) $\varepsilon_n^2 \|x\|^2 \leq \delta_n^2$ for all $n \in \mathbb{N}$ implies that $\|x\| \leq r$. $\square$

Now we can finish the proof of Theorem 1. For each component of W_{δ_n} there is a component of the boundary H_{δ_n} ; thus W_{δ_n} has at most $d(2d-1)^{n-1}$ components. Then the intersection $\bigcap_{n \in \mathbb{N}} W_{\delta_n}$ also has at most $d(2d-1)^{n-1}$ connected components since for large n , W_{δ_n} is contained in an arbitrarily small neighborhood of the intersection. $\square$

Theorem 1 can be used to bound the number of connected components of semi-algebraic sets. We give a first bound in the next proposition that is useful in Section 16.2. A finer bound is proved in Section 16.3.

Proposition 3 Suppose $S \subseteq \mathbb{R}^n$ is defined by

$$\begin{cases} f_i(x) = 0, & i = 1, \dots, p \\ f_i(x) \geq 0, & i = p+1, \dots, p+k \\ f_i(x) > 0, & i = p+k+1, \dots, p+\ell, \end{cases}$$

and let $d = \max\{\text{degree } f_1, \dots, \text{degree } f_{p+\ell}, 2\}$. Then the number of connected components of S is bounded by $d(2d-1)^{n+\ell-1}$.

Proof. Our first goal is to replace the strict inequalities by nonstrict ones. To do so, let j be the number of connected components of S and $v_1, \dots, v_j$ be points in distinct connected components of S . Consider

$$\varepsilon = \min_{\substack{i=1, \dots, \ell-k \\ s=1, \dots, j}} f_{p+k+i}(v_s)$$

and S' the subset of $\mathbb{R}^n$ defined by

$$\begin{cases} f_i(x) = 0, & i = 1, \dots, p \\ f_i(x) \geq 0, & i = p+1, \dots, p+k \\ \widehat{f}_i(x) = f_i(x) - \varepsilon \geq 0, & i = p+k+1, \dots, p+\ell. \end{cases}$$

Then S' has a point in each of the j connected components of S and $S' \subset S$. Thus S' has at least j connected components. Now, for each inequality

$$\begin{aligned} f_i(x) &\geq 0 \quad i = p+1, \dots, p+k \\ \widehat{f}_i(x) &\geq 0 \quad i = p+k+1, \dots, p+\ell, \end{aligned}$$

add a new variable y_i and replace

$$\begin{aligned} f_i(x) &\geq 0 \text{ by } f_i(x) - y_i^2 = 0 \\ \text{and } \widehat{f}_i(x) &\geq 0 \text{ by } \widehat{f}_i(x) - y_i^2 = 0. \end{aligned}$$

Finally, let $V \subset \mathbb{R}^{n+\ell}$ be defined by the equations

$$\begin{cases} f_i(x) = 0, & i = 1, \dots, p \\ f_i(x) + y_i^2 = 0, & i = p+1, \dots, p+k \\ \widehat{f}_i(x) + y_i^2 = 0, & i = p+k+1, \dots, p+\ell. \end{cases}$$

Since the degree of the polynomials defining V is at most d , we apply Theorem 1 and get an upper bound of $d(2d-1)^{n+\ell-1}$ for the number of its connected components. But S' is the projection of V on $\mathbb{R}^n$ and therefore the same bound holds for the number of connected components of S' and a fortiori for that of S . $\square$

16.2 A Complexity Lower Bound

In this section we use the upper bounds computed in the last section to get lower bounds for the depth of algebraic decision trees.

Definition 1 An algebraic computation tree is a finite-dimensional machine whose graph of nodes is acyclic and whose computations are arithmetic operations. That is, a computation node performs a single arithmetic operation $x \circ y$, where $\circ$ may be $+$, $-$, $\times$, or $/$, and x and y are the contents of certain machine registers or the machine constants. The number of nodes of the machine is the *size* of the tree and the length of the path from the *root* (input node) to a given node is the *depth* of the node. The largest depth of an output node is the *depth* of the tree. The output nodes of the tree are called *leaves*.

An algebraic decision tree is an algebraic computation tree in which each output node is labeled Yes or No. If T is an algebraic decision tree with input space $\mathbb{R}^n$ we say that T *accepts* $x \in \mathbb{R}^n$ if the computation of T with input x leads to a leaf labeled Yes. Otherwise we say that T *rejects* x .

Remark 1 If T is an algebraic computation tree with depth d , the number of its leaves is bounded by 2^d . This follows from the fact that each node of T has at most two next nodes.

Remark 2 In Part III, we deal mainly with decision problems and thus, our machines often compute characteristic functions. We adopt for these machines the terminology used for algebraic decision trees. That is, for a given input x , we say that machine M *accepts* x if $\Phi_M(x) = 1$ and that it *rejects* x otherwise.

Theorem 2 Let $W \subseteq \mathbb{R}^n$ be any set and let T be an algebraic decision tree that decides W . Then the depth t of T satisfies the bounds

$$(i) \quad t \geq \frac{1}{(1 + \log 3)} \log H - \frac{\log 3}{(1 + \log 3)} n \geq 0.38 \log H - 0.61n,$$

$$(ii) \quad t \geq \frac{1}{(1 + 3 \log 3)} \log H - \frac{1 + \log 3}{1 + 3 \log 3} n \geq 0.21 \log H - 0.55n,$$

where H is the number of connected components of W .

Proof. Let $x \in \mathbb{R}^n$ and let $\gamma_x = (v_1, \dots, v_h)$, $h \leq t$ be the path traversed by x from the root v_1 of T to a leaf v_h . Let $\mathcal{V}_x \subseteq \mathbb{R}^n$ be the set of points in $\mathbb{R}^n$ that follow this same path. We now bound the number of connected components of $\mathcal{V}_x$ by describing it succinctly by a system of quadratic equations and inequalities.

To each computation node that is not a division node v_i add a new variable y_i and the equation $y_i = z_k \circ z_\ell$, where $z_k, z_\ell \in \{x_1, \dots, x_n, y_1, \dots, y_{i-1}\}$ and $\circ$ is the operation performed at v_i . If v_i is a division node, add the variable y_i and equation $y_i z_\ell = z_k$, where z_ℓ, z_k are as above and the division performed at v_i is z_k/z_ℓ .

If v_i is a branch node, add the inequality $z_k \geq 0$ if $v_{i+1} = \beta^+(v_i)$ and $z_k < 0$ otherwise, where $z_k \in \{x_1, \dots, x_n, y_1, \dots, y_{i-1}\}$ is the value whose sign is tested.

Let $S \subseteq \mathbb{R}^{n+h-1}$ be the semi-algebraic set defined by the system of quadratic equations and inequalities previously described. By Proposition 3 the number of connected components of S is bounded by $2(3)^{n+h-1}$ and therefore by $2(3)^{n+t-1}$. Since $\mathcal{V}_x$ is the projection onto the first n coordinates of S we deduce that the same bound holds for the number of connected components of $\mathcal{V}_x$.

The set W is the union of the $\mathcal{V}_x$ for all paths γ_x ending in a leaf labeled yes. Since there are at most 2^t such leaves, we deduce that the number of connected components of W is bounded by $2^{t+1} 3^{n+t-1}$. The lower bound for t follows, thus proving (i).

To prove (ii) let r denote the number of coordinates of $(x_1, \dots, x_n)$ that occur in the equalities and inequalities describing $\mathcal{V}_x$. Then $r \leq n$, and the preceding reasoning shows that the number of connected components of $\mathcal{V}_x$ is bounded by $2(3)^{r+t+1}$. But $r \leq 2t$ since at each node in $\mathcal{V}_x$ at most two new coordinates can appear. Thus, the number of connected components of $\mathcal{V}_x$ is at most $2(3)^{3t+1}$ and consequently the number of connected components of W is at most $2^{t+1} 3^{3t+1}$ from which (ii) follows. $\square$

Theorem 2 yields nonuniform lower bounds since it is stated in terms of algebraic decision trees. To deduce from it lower bounds for uniform machines the following proposition is helpful.

Proposition 4 Let M be a uniform machine with halting time bounded by a function $t : \mathbb{N} \rightarrow \mathbb{N}$. For each $n \in \mathbb{N}$ there is an algebraic computation tree $T_{M,n}$ computing the restriction of φ_M to $\mathbb{R}^n$. If M decides a set $S \subseteq \mathbb{R}^\infty$, then $T_{M,n}$ can be considered to be an algebraic decision tree and the set decided by it is $S_n = S \cap \mathbb{R}^n$.

Proof. We have seen in Proposition 1 of Chapter 3 that for any input size n there is a finite-dimensional machine M_n computing the restriction of φ_M to $\mathbb{R}^n$ in time

at most $t(n)$. A tree T_1 is obtained from M_n by unfolding its graph up to length $t(n)$.

We now remove from T_1 all the subtrees whose leaves are not output nodes. Since M_n works in time at most $t(n)$ we know that no $x \in \mathbb{R}^n$ will reach one of these leaves. Let T_2 be the resulting tree. The tree $T_{M,n}$ is obtained from T_2 by replacing each computation node by a sequence of arithmetic operations computing the rational function associated with the node. This increases the depth of $T_{M,n}$ with respect to the depth of T_2 by only a constant factor.

If M is a decision machine, an additional step is needed. We replace each leaf by a branching node with the test $x_1 = 1$ followed by a Yes leaf corresponding to the answer $x_1 = 1$ and a No leaf corresponding to the answer $x_1 \neq 1$. $\square$

Remark 3 Theorem 2 gives a lower bound with precise constants for the depth of algebraic decision trees deciding a given set $W \subset \mathbb{R}^n$. In passing to uniform machines, this precision gets lost for two reasons: since the computation nodes of uniform machines can be arbitrary rational functions, there is a possible speedup by a constant factor; and, management introduces extra computing time that does not affect the lower bound but makes the constant multiplying of the $\log H$ meaningless.

These considerations suggest that, in the case of uniform machines, Theorem 2 will be used mainly to deduce the order of magnitude of lower bounds for the halting time. We now introduce notation that plays a role for lower bounds closely related to the "big Oh" notation introduced in Section 4.4 for upper bounds.

Definition 2 Let $f, g : \mathbb{N} \rightarrow \mathbb{R}$. We say that f is $\Omega(g)$ if there are $n_0 \in \mathbb{N}$, $c \in \mathbb{R}$, $c > 0$ such that for all $n \geq n_0$, $|f(n)| \geq c|g(n)|$.

Theorem 3 Let $S \subseteq \mathbb{R}^\infty$ and let S_n be its subset of elements having size n . Then the halting time of any machine M deciding S is $\Omega(\log g(n))$ where $g(n)$ is the number of connected components of S_n . $\square$

Remark 4 In this theorem, the constant hidden in $\Omega(\log g(n))$ depends on M . If we restrict to machines whose computation nodes only perform arithmetic operations, then this constant can be taken to be independent of the machine.

We now give some applications of Theorems 2 and 3. In each of these, we want to estimate the number of connected components of certain sets. Since these components are open subsets of $\mathbb{R}^n$ they are also path connected. We use this fact freely.

Example 2 Let us consider the problem of deciding whether a point x in $\mathbb{R}^\infty$ has two different coordinates $i, j \leq \text{size}(x)$ such that $x_i = x_j$. Thus, we want to decide the set

$$S = \{x \in \mathbb{R}^\infty \mid x_i = x_j \text{ for } i, j \leq \text{size}(x), i \neq j\}.$$

We denote by W_n the set of points having size n that do not belong to S . Then $W_n \subset \mathbb{R}^n = \{(x_1, \dots, x_n) \mid x_i \neq x_j \text{ for } i \neq j\}$.

Now, take the point $(1, 2, \dots, n) \in W_n$ and consider points $x, y \in W_n$ given by two different permutations to its n coordinates.

There exist coordinates $i, j \leq n$ such that $x_i - x_j > 0$ and $y_i - y_j < 0$. Therefore, for any path $\gamma : [0, 1] \rightarrow W_n$ connecting x and y with $\gamma(0) = x$ and $\gamma(1) = y$, one has that the composition

$$g = (x_i - x_j) \circ \gamma : [0, 1] \rightarrow \mathbb{R}$$

sends 0 to a positive number and 1 to a negative one. By the intermediate value theorem, there is a point $c \in (0, 1)$ such that $g(c) = 0$. But then, $\gamma(c)$ has equal coordinates i and j and cannot belong to W_n . One deduces that x and y are not path connected and that the number of connected components of W_n is at least $n!$.

Since the union of the W_n is the complement of S , a direct application of Theorem 3 yields a lower bound of $\Omega(n \log n)$ for the halting time of any machine deciding S .

Example 3 In much the same manner, one shows that the problem of deciding, given $x_1, \dots, x_n, y_1, \dots, y_n \in \mathbb{R}$, whether $y_i = x_j$ for some $i, j \leq n$, also has a lower bound of $\Omega(n \log n)$.

We close this section with an example that has been widely studied in algebraic complexity.

Definition 3 Let $V \subset \mathbb{C}^n$ be an algebraic set defined by $f_1 = 0, \dots, f_k = 0$, and let r be its dimension. Let L be an affine subspace of dimension $n - r$ and $\#(L \cap V)$ be the cardinality of $L \cap V$. The *degree* of V is the maximum of $\#(L \cap V)$ over all affine subspaces with $\#(L \cap V) < \infty$.

Example 4 Let $V \subseteq \mathbb{C}^n$ be as in the preceding definition and T be an algebraic decision tree that decides V . We can assume that T takes its inputs from $\mathbb{R}^{2n}$. Now let L be an affine $(n - r)$ -dimensional subspace such that $\#(L \cap V) = \text{degree } V$. With an additional $2rn$ operations one can decide whether the input belongs to L and thus to $L \cap V$. Therefore, if t is the depth of T one deduces from Theorem 2 that

$$t + 2rn \geq 0.38 \log(\text{degree } V) - 1.24n$$

and finally that

$$t \geq 0.38 \log(\text{degree } V) - (1.24 + 2r)n.$$

16.3 On the Number of Connected Components of Semi-Algebraic Sets

In this section we use Theorem 1 to bound the number of connected components of semi-algebraic sets. The bound obtained depends exponentially only on the number of variables, thus refining Proposition 3. On the way, we also obtain a bound for

the number of sign conditions that can be satisfied by a set of polynomials. Both results are used in later chapters.

We begin with a particular case.

Proposition 5 Let $S \subset \mathbb{R}^n$ be defined by

$$\begin{cases} f_i(x) = 0, & i = 1, \dots, p \\ f_i(x) > 0, & i = p+1, \dots, p+s \end{cases}$$

and let $d = \max\{\text{degree } f_1, \dots, \text{degree } f_{p+s}\}$. Then the number of connected components of S is bounded by $(sd + 1)(2sd + 1)^n$.

Proof. Let $W = \{x \in \mathbb{R}^n \mid f_i(x) = 0, i = 1, \dots, p \text{ and } f_i(x) \neq 0, i = p+1, \dots, p+s\}$. Each connected component of S is a connected component of W . Therefore the number of connected components of S is at most that of W . But W is the projection on $\mathbb{R}^n$ of the algebraic set in $\mathbb{R}^{n+1}$ given by the polynomial equalities in $(x_1, \dots, x_n, y)$,

$$f_1(x_1, \dots, x_n) = 0, \dots, f_p(x_1, \dots, x_n) = 0, y \prod_{i=p+1}^{p+s} f_i(x_1, \dots, x_n) - 1 = 0,$$

that has, according to Theorem 1, at most $(sd + 1)(2sd + 1)^n$ connected components. $\square$

Definition 4 A *sign condition* is any of the three elements in $\{<, =, >\}$. A polynomial $f \in \mathbb{R}[X_1, \dots, X_n]$ satisfies a sign condition σ if there is an $x \in \mathbb{R}^n$ such that $f(x)\sigma 0$.

For a set $\{f_1, \dots, f_k\}$ of polynomials in $\mathbb{R}[X_1, \dots, X_n]$ and a tuple of sign conditions $\sigma = (\sigma_1, \dots, \sigma_k)$, we say that the set satisfies σ if there is an $x \in \mathbb{R}^n$ such that $f_i(x)\sigma_i 0$ for all $i = 1, \dots, k$. If the set of polynomials is clear from the context, we say that σ is *satisfiable*.

Proposition 6 Let $f_i \in \mathbb{R}[X_1, \dots, X_n]$ be a polynomial of degree d_i for $i = 1, \dots, k$ and let $D = \sum_{i=1}^k d_i$. The number of tuples of sign conditions satisfied by the set $\{f_1, \dots, f_k\}$ is at most $(2D + 1)(4D + 1)^n$.

Proof. Let $A \subset \mathbb{R}^n$ be a set containing exactly one point in each nonempty set defined by a tuple of sign conditions. Thus, the cardinality of A is the number we want to bound. We denote by A'_i the subset of A containing those points on which the sign of f_i is either $>$ or $<$. We then choose $\delta > 0$ such that for all $i \leq k$ and all

$x \in A'_i$ one has $|f_i(x)| > \delta$ and consider the polynomial $\tilde{f} = \prod_{i=1}^k (f_i - \delta)(f_i + \delta)$

whose zero set is the dotted lines in Figure C.

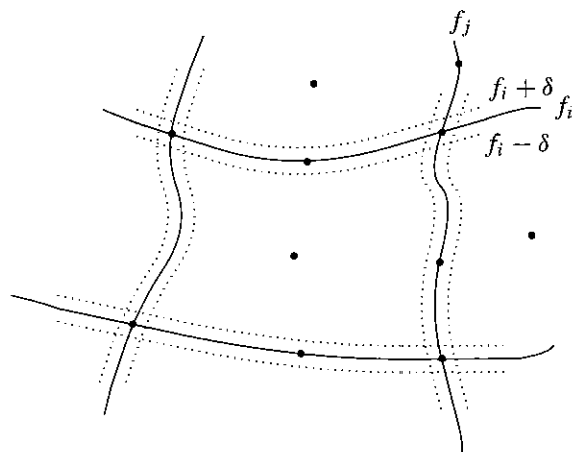


Figure C The zero set of $\tilde{f}$.

We claim that each connected component of W , the complement of the zero set of $\tilde{f}$, contains at most one point of A . To prove this claim, consider two different points x, y in A . Then there exists $i \leq k$ such that, for instance, $f_i(x) > 0$ and $f_i(y) \leq 0$. Because of the choice of δ we then have that $f_i(x) - \delta > 0$ and since $\delta > 0$ that $f_i(y) - \delta < 0$. If x and y were in the same connected component of W , then it would be a path γ joining them inside W . But as in Example 2, any such path crosses the set of zeros of $f_i - \delta$ and therefore cannot be contained in W .

Finally, the number of connected components of $\tilde{f} \neq 0$ is the same as the number of connected components in $\mathbb{R}^{n+1}$ of the zero set of the polynomial $Y\tilde{f} - 1$. Applying Theorem 1 to this polynomial we deduce the statement of the proposition. $\square$

Proposition 7 Let $S \subseteq \mathbb{R}^n$ be defined by

$$\begin{cases} f_i(x) = 0, & i = 1, \dots, p \\ f_i(x) \geq 0, & i = p+1, \dots, p+\ell \\ f_i(x) > 0, & i = p+\ell+1, \dots, p+\ell+k \end{cases}$$

and let $d = \max\{\text{degree } f_1, \dots, \text{degree } f_k\}$. Then the number of connected components of S is bounded by $(kd+1)(2kd+1)^{n+1}(4kd+1)^n$.

Proof. The set S is the union of sets S_j given by inequalities of the form $f_i \sigma_i 0$ for $i = 1, \dots, k$ where $\sigma_i \in \{=, >\}$. By Proposition 6 there are at most $(2kd+1)(4kd+1)^n$ nonempty sets S_j . And by Proposition 5, each has at most $(kd+1)(2kd+1)^n$ connected components. Multiplying both bounds finishes the proof. $\square$

16.4 Additional Comments and Bibliographical Remarks

The bounds on the number of connected components of algebraic sets established in Section 16.1 are due to Oleinik and Petrovski [1949] with an important early special case done by Petrovski (see also [Oleinik 1951]) and, independently, to Milnor [1964] and Thom [1965].

The use of the number of connected components to provide complexity lower bounds originates with Dobkin and Lipton [1979] who considered linear trees, that is, trees performing only linear operations. To the best of our knowledge, the first use of the bounds obtained in Section 16.1 is due to Steele and Yao [1982]. Part (i) of Theorem 2 is due to Ben-Or [1983]; its modification in Part (ii) is taken from [Cucker, Montaña, and Pardo 1992].

There are several results providing bounds for the number of satisfiable sign conditions of a polynomial family. A classic reference is [Warren 1968]. Proposition 6 is taken from [Meyer auf der Heide 1985b].